

Sigil: Stratified Invariant Generative-Image Lineage

A watermark whose false-positive rate is a design parameter,
and whose robustness is spread across strata that fail differently

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Abstract

Image watermarking is usually evaluated as a robustness contest and reported as a single true-positive rate. That framing hides the two things that decide whether a provenance mark is usable: what its false-positive rate actually is, and which attacks it is structurally unable to survive. SIGIL is built around both.

The detector’s null distribution is not fitted. Every statistic SIGIL reports is a keyed Rademacher sum, so its tail is bounded exactly by Hoeffding’s inequality for every image, every finite carrier count, and no distributional assumption at all; the searches the detector performs over anchors, scales, rotations and nonces are charged for explicitly — including the refinement the detector opens around whichever hypothesis the data selects — and the strata are combined by a weighted Bonferroni rule that is valid under arbitrary dependence. The false-positive rate is therefore a number the operator sets rather than a number the experiment measures. The core of that argument — the exact tail, the multiplicity charge, the fusion, and the two invariances the analytic stratum rests on — is machine-checked in Lean 4 against Mathlib.

Robustness comes from stratification rather than from a single clever transform. An analytic stratum places a multiplicative mark in the ring-normalised log-magnitude of a canonical spectrum, which makes it exactly invariant to translation, to any zero-phase radially symmetric filter, and to global gain, and gives it high payload capacity at negligible cost. A learned stratum is trained with real variational autoencoders and diffusion regeneration inside its noise layer, together with a jointly-trained removal network and a white-box gradient attack, so it reads through exactly the transformation that erases every linear spectral mark. It is carried at two message-grid scales, which trade crop robustness against rotation robustness through a single parameter in opposite directions, so that fusing them covers a geometric range neither reaches alone. Carriers in every stratum are bound to the image — by content descriptors and by a per-image nonce — so an adversary who averages the residuals of an arbitrarily large corpus recovers little. Measured as the attack itself — aggregate the residuals, rank the bins, take the top K — a static-

carrier control surrenders 50.0% of its carrier set, while content-derived carriers yield 9.5% against a chance rate of 0.74%.

Evaluated on — images across — attacks — valuemetric, geometric, codebook and collusion, generative regeneration, white-box adaptive, and stacked pipelines — under an explicit attacker quality budget, SIGIL holds at a measured false-positive rate of — % while marking at — dB PSNR and — SSIM.

1 Introduction

A provenance watermark makes a claim about an image: this came from a particular generator, at a particular time, under a particular key. The claim is only worth as much as the rate at which it is made about images to which it does not apply. That rate — the false-positive rate — is the quantity a deployment must be able to bound, because the cost of a false accusation is not symmetric with the cost of a miss.

Most published watermarking systems fit that rate. A detection statistic is computed on a corpus of unmarked images, a Gaussian or empirical null is fitted, and a threshold is read off. Three things go wrong. The fitted null is a property of the corpus, and an adversary is not obliged to supply images from it. The detector usually maximises its statistic over a synchronisation search — grid offsets, rotations, scales — and the maximum of many draws is not distributed like a single draw. And when a system reports several detectors and takes the best, the same inflation happens again at the top level. A reported “0% false positives on our test set” is consistent with a real rate orders of magnitude higher.

SIGIL takes the opposite route. Every statistic it computes has the form $T = \sum_k b_k z_k / \|z\|$, where the evidence z is read from the image and the signs b come from a keyed pseudo-random function. Under the null — this image was never marked with this key — the image is independent of the key, so conditionally on z the statistic is a normalised Rademacher sum with known weights, and

$$\mathbb{P}[T \geq t] \leq e^{-t^2/2}$$

holds exactly, for every image, at every finite carrier count, with no assumption whatsoever about image

statistics. Searches are charged for by an explicit union bound and strata are combined by weighted Bonferroni. The result is a detector whose false-positive rate is a parameter rather than a measurement (§4).

The second half of the design is about what no single mechanism can do. A mark that lives in a linear transform of the pixels can be made exactly invariant to translation, to radially symmetric filtering, and to contrast — but a variational autoencoder round-trip rewrites the mid-band spectrum wholesale, and no imperceptible linear perturbation of that band survives it. A mark that a network is trained to read through such a round-trip survives it — but the network has no invariance anyone can prove and no natural exact null. SIGIL carries both and spends one false-positive budget across them (§3).

Binding the carriers to the image is what defeats the attack that breaks static designs. If the carrier set comes from the key alone, an adversary who aggregates the residuals of many marked images sees the same bins light up every time, learns the carrier set without learning the key, and erases the mark by whitening exactly those bins — at a distortion cost no larger than the embedder paid. SIGIL keys its carriers to three anchors of different character: two robust content descriptors, and a per-image random nonce recovered from the learned stratum. The nonce matters because content descriptors are clusterable: an adversary who can group images by appearance can group images by descriptor, and run the codebook attack inside a group. A nonce has entropy that does not depend on content at all (§3.3).

Contributions.

1. A watermark detector whose false-positive rate is exact, finite-sample and distribution-free, including the cost of its own synchronisation search and of fusing several detectors (§4).
2. A machine-checked development in Lean 4 of the four results the security argument depends on: the exact null, the multiplicity charge, the fusion rule, and the two invariances of the analytic stratum (§4.1).
3. An analytic stratum in the ring-normalised log-magnitude domain that is exactly invariant to translation, radially symmetric zero-phase filtering and global gain, and that resynchronises crops and reflections by resampling a single spectrum rather than re-rendering the image (§3.2).
4. Multi-anchor content keying, including cross-stratum nonce anchoring, with measured diversity and per-attack stability for each anchor family (§3.3).
5. An evaluation protocol with an explicit attacker quality budget, a wrong-key false-positive control, and — attacks spanning six families including white-box gradient removal against the deployed decoder (§5).

2 Threat model and protocol

What the adversary knows. Everything except the key: the architecture, the training data, the network weights, the descriptor definitions, the quantiser steps, the search grids, and the fact that an image is marked. For the adaptive attacks in §5 the adversary also holds the transmitted message and takes gradients through the deployed decoder.

What the adversary wants. An image that a detector will not flag and that a viewer would still accept in place of the original. Both halves matter. A transform that drives PSNR to 15 dB has not removed the watermark; it has removed the picture, and an adversary willing to do that could have posted noise instead.

Definition 1 (Quality budget). An attack is *admissible* at budget (π, σ) if the attacked image satisfies $\text{PSNR} \geq \pi$ and $\text{SSIM} \geq \sigma$ against the marked image, *except* for two families that a pixelwise metric cannot describe, which are admissible by construction: geometric transforms — translation, crop, rescale, rotation, reflection — which move pixels rather than damage them, and global photometric edits — gamma, brightness, contrast, saturation — which a viewer reads as a colour grade.

Remark 2. This exemption runs against the system’s own interest and is the reason it is stated explicitly. A seven-pixel cyclic shift scores 13 dB PSNR and 0.14 SSIM against the unshifted original while being visibly the same picture; a gamma correction scores about 20 dB. Judging those by PSNR would let a watermark claim robustness by *disqualifying* the attacks a real adversary would reach for first. The cost of these attacks is paid in what the detector has to resynchronise or absorb, and it is reported there.

The exemption needs one guard, because a stacked attack can otherwise inherit it for free. A pipeline whose first step is a generative regeneration and whose second is a crop reports no finite PSNR at all — the crop has moved the pixels — and would be admitted whatever the regeneration did to the picture. So a stack is scored on the part of it that did not move: drop the geometric steps, apply the rest to the original, and require *that* chain to meet the budget. It is alignment-preserving, so the metric means what it usually means, and the geometry keeps its exemption without extending it to everything stacked beside it.

Remark 3. The guard is not free to SIGIL. It disqualifies `img2img→crop→jpeg`, a stack that defeats SIGIL outright — but it disqualifies it because its regeneration step alone scores 0.59 SSIM, below the budget that already ruled the same regeneration inadmissible when run on its own. Admitting the stack while excluding its

own first step would have been an inconsistency, not a concession. The stack’s result is reported in full in §5 regardless, as are all three systems’ results on it, which are identical: none survives it.

Every attack in §5 reports its own distortion and is tagged against the budget. Results outside the budget are reported — they mark where the architecture ends — but are not counted as defeats.

The two false-positive controls. Reporting a detection rate without a matching false-positive rate is meaningless, and the usual control — run the detector on unmarked images — is the weaker of the two available. SIGIL also reports a *wrong-key* control: the same image, marked under a key the detector does not hold. A detector that fires there is reacting to the act of embedding rather than to the key, and every positive it produces is worthless.

3 Architecture

3.1 Canonicalisation and residual transport

Both strata work on a canonical grid and push their result back to native resolution. Given an image of shape $H \times W$, the canonical grid is the aspect-preserving shape of area N^2 ($N = 512$ analytic, 256 learned):

$$H_c = N\sqrt{H/W}, \quad W_c = N\sqrt{W/H}.$$

Preserving the aspect ratio is not cosmetic. A fixed square grid makes canonical pixels non-square for every non-square image, and an image-plane rotation then becomes a shear of the canonical spectrum rather than a rotation of it — which destroys the resynchronisation the analytic stratum depends on.

A stratum produces a canonical-grid luminance residual; the residual is resampled to $H \times W$ and added to the luminance channel, leaving chrominance untouched. Because every residual is band-limited well below the canonical Nyquist rate the resampling is faithful, and the immediate consequence is that a uniform rescale of the delivered image maps back onto the same canonical representation. Scale invariance is a property of the representation, not something the detector searches for.

3.2 The analytic stratum

Let x be the canonical luminance, $F = \mathcal{F}x$ its DFT, and let the bins be partitioned into rings $R(\rho)$ of equal radial frequency. The detector’s evidence at bin f is the ring-standardised log-magnitude

$$z(f) = \frac{\log |F(f)| - \mu_{R(f)}}{\sigma_{R(f)}}, \quad (1)$$

with μ_R, σ_R the mean and standard deviation of $\log |F|$ over the ring. Subtracting the ring mean is what delivers the invariances of §4; dividing by the ring standard deviation costs nothing and recovers real detection power, because the natural spread of $\log |F|$ is much larger at low radii than at high ones and an unstandardised statistic is dominated by its noisiest bins.

Embedding is a *proportional* modulation. For each carrier bin f_k with keyed chip $b_k \in \{\pm 1\}$ and strength α_k ,

$$F(f_k) \leftarrow F(f_k) e^{b_k \alpha_k}, \quad (2)$$

with the conjugate bin given the same log-gain so the modified spectrum stays Hermitian and the inverse transform is exactly real. Proportionality earns its place twice over. It follows Weber’s law, so the perturbation hides inside whatever energy the image already has at that bin. And it is exactly sign-symmetric, so it leaves every band-averaged spectral statistic unbiased — which matters because one of SIGIL’s own content anchors is a band-averaged spectral statistic, and an absolute rule (forcing bins to $\mu_R \pm \alpha$) would systematically lift low-magnitude bins and let the mark perturb the anchor that selects it.

The detector correlates z at the carrier positions against the chips:

$$T = \frac{\sum_k b_k z(f_k)}{(\sum_k z(f_k)^2)^{1/2}}. \quad (3)$$

Per-bin evidence is soft-clipped at a robust multiple of its own median absolute deviation before (3), which stops a handful of outlier bins — spectral leakage from a specular highlight, a JPEG blocking harmonic — from inflating the denominator. The clip is a function of the observed image only, never of the keyed signs, so the exact null of §4 is untouched.

Resynchronisation. Rotation, crop and reflection act on the canonical spectrum as a rotation, a radial dilation and an axis flip. The detector does not re-render the image for any of these: it resamples the already-computed field z at transformed carrier coordinates, so a hypothesis costs a few thousand bilinear taps. The grid must be fine — a carrier at radius r moves by $r \delta s$ bins under a scale error, so half-bin accuracy at the outer radius needs a few hundred scales — which is why the search runs in two stages: a proposal pass over a carrier subset, then a full-carrier pass on a local neighbourhood of each surviving proposal. The staging is a compute optimisation only; the reported p -value always pays Bonferroni for the entire nominal grid.

3.3 Content anchors

Carrier positions and chip signs are drawn from a keyed PRF applied to an *anchor*. A good anchor is robust, be-

cause the detector re-derives it from an attacked image, and high-entropy, because an adversary who can group images into a few hundred equivalence classes can run the codebook attack inside a class. No single descriptor is both across all attacks, so SIGIL carries three on disjoint carrier subsets and detects on whichever survives:

spectral the ripple of the low-band radial log-power profile, detrended against a quadratic in $\log r$. Any smooth radial filter adds a smooth function of $\log r$ to that profile, so removing the polynomial subspace gives the anchor the same filtering invariance the detector statistic has. It slides bodily under a crop and does not survive rotation.

histogram order statistics of the luminance and chroma marginals, normalised by their own interquartile range. A crop, a rotation or a rescale resamples the same scene and barely moves a distribution’s quantiles; heavy blur does.

nonce a per-image uniform value, recovered not from the pixels but from the learned stratum. Its entropy is independent of content, so an adversary who clusters images by appearance learns nothing about it. This is the anchor that closes the clustering loophole the two content descriptors leave open.

Every anchor is *list-decoded*: a quantiser cell boundary is the only place a content anchor is fragile, so the detector never commits to one cell but walks outward along the coordinates whose residual sits closest to a boundary. Besides absorbing attack-induced drift, this disarms the adversary who deliberately nudges an image across a boundary — the cell they push it into is already on the list. Measured diversity and per-attack stability are in Table 3.

3.4 The learned stratum

An encoder renders a message as an additive canonical-grid residual and a decoder reads it back. Three choices carry the design.

A learned spread-spectrum code. The message is rendered by a bank of learned chips — one 64×64 tile per bit, summed with that bit’s sign and tiled across the canonical grid — fed both into a U-Net and directly to its output layer. Routing the message instead through a learned bottleneck (a dense layer into a small grid, then upsampling convolutions) does not train at all: encoder and decoder must co-adapt through a mapping that starts random, and the message loss sits at $\ln 2$ indefinitely. The chip bank puts a plain spread-spectrum solution one linear map away on the first step, and training is then free to shape it into something the attacks cannot remove.

Periodic coding and per-location logits. The chip field is spatially periodic and the decoder predicts the whole message at every spatial location, averaging *logits* rather than pooling features. Each location is an independent estimate of the same quantity; averaging estimates is far easier to learn than decoding one pooled summary, and it is exactly what makes the decoder translation-invariant and therefore crop-tolerant. The decoder’s first stages run at full resolution: striding immediately aliases away the very perturbation it is meant to read, because early in training the encoder’s mark is high-frequency and a strided convolution discards it before any gradient can teach the encoder to move somewhere safer.

Nonce-keyed messages. The transmitted message is $m(\nu)$, the codeword of a per-image random nonce under a keyed linear code over $\text{GF}(2)$. A fixed codeword would be a static codebook for the learned stratum, recoverable by averaging residuals. Blind detection must then find the nonce, which naively means correlating against all 2^k codewords; because the code is linear, that is a Walsh–Hadamard transform — scatter the logits into bins indexed by the generator rows, transform once, and every codeword correlation appears at once in $O(2^k k)$ time. A 24-bit nonce space costs tens of milliseconds, and the p -value simply pays Bonferroni for the 2^k hypotheses it searched.

Training. The noise layer runs the *real* operation forward and a straight-through surrogate backward whenever an operation is not differentiable, so there is no gap between the JPEG the encoder trains against and the JPEG it is tested on. It contains real variational autoencoders — two different ones, so the encoder cannot overfit one decoder’s reconstruction quirks — with latent noise, which is what a low-strength `img2img` sampler actually does to the code and is the step that destroys spectral marks. From a fixed schedule point onward a jointly-trained removal network and a short white-box PGD attack on the decoder are mixed in, so robustness is measured against an opponent that searches rather than against a fixed menu. Attack severity is annealed from near-identity: a cold encoder emits a residual of exactly zero, and if the first batches are cropped to a third and rotated 25° there is no signal to grow a mark from.

3.5 Two message grids

The learned stratum’s message is positional: bit k lives in cell k of a $g \times g$ grid over the canonical frame, so a cell is $256/g$ canonical pixels across. That single number sets both geometric limits, and it sets them in opposite directions.

A centre crop discards whichever cells fall outside the surviving frame, so what matters is how many remain. A statistic saturates at $\sqrt{n}$ in the number of readable bits, so surviving a crop to a fraction f of the frame needs $f^2 g^2$ to stay comfortably above the threshold squared — which argues for many small cells.

A rotation argues the other way. The detector cannot read a rotated positional code at all; it resynchronises, applying the inverse rotation and then decoding. The damage it must survive is therefore two bilinear resamplings, each blurring by about one canonical pixel, and that fixed blur eats a fraction of a cell that grows as the cell shrinks — which argues for few large cells.

SIGIL does not choose. It carries two learned strata, a fine grid at 32×32 and a coarse grid at 16×16 , each embedded at a lower amplitude so their residuals sum to the fidelity a single stratum would have spent alone. Measured at matched training steps, the trade is exactly as the argument predicts and the curves cross (Figure 4): the fine grid carries twice the clean evidence and holds the tighter crops, while the coarse grid holds the larger rotations, and each is the stronger of the two precisely where the other has failed.

The two grids share a fixed residual-energy budget. Moving energy between them over a two-to-one range leaves the measured crop and rotation rates nearly unchanged, which shows that complementarity comes from geometry rather than volume. The fine grid supplies more cells for crop evidence; the coarse grid places the message at lower spatial frequency and therefore retains more detail under heavy quantisation. In the measured suite, quality 20 JPEG improves from 60% to 100% with the coarse grid, while the fifteen-degree rotation rate changes from 21% to 32%. The same cell-size parameter controls both geometric resynchronisation and sensitivity to detail-removing attacks.

3.6 Fusion

Each stratum reports a p -value already valid for everything it searched. The fused decision is a weighted Bonferroni rule,

$$p = \min_i \frac{p_i}{w_i}, \quad \sum_i w_i \leq 1,$$

over the active analytic and learned strata, which requires no independence — and that is not a technicality, since the adversary chooses the image and can deliberately correlate what the strata see. Detection is $p \leq \alpha$. The weights are equal thirds: which stratum survives a given attack is not known before the attack, so weighting one above another would encode a guess about the adversary rather than a property of the design.

Carrying a third stratum costs one more term in that minimum, which is to say a factor of three-halves in the budget each stratum receives, worth $\sqrt{2 \log(3/2)} \approx 0.9$

on the threshold. It buys a geometric range no single grid achieves.

Remark 4 (Why not pool the evidence?). Bonferroni is valid under arbitrary dependence, which is more than the two learned grids need: their chips are drawn under different domain-separation labels, so conditionally on the image their statistics are independent, and a pooling rule such as Fisher’s is available. It is also worse here. Fisher spends its budget on the *sum* of the two log p -values, so when one stratum is at chance the other must reach $p \leq 1.2 \times 10^{-9}$ to clear a χ_4^2 threshold, against 3.3×10^{-7} for the minimum rule — pooling helps only when both strata carry partial evidence at once. Replaying the whole corpus through both rules, Fisher changes five conditions of sixty-five and loses on every one of them, including the rotation it was tried for (32.5% $\rightarrow$ 28.7%); the other sixty are untouched.

That is not an accident of these numbers. SIGIL is built so that its strata fail on disjoint attacks, so the case where one carries alone is the design working, not an edge case — and it is exactly the case pooling handles badly. A rule that assumes less about the dependence turns out to assume the right thing about the architecture.

4 Theory

Assumption 5 (Keyed PRF). The key-derivation function behaves as a random oracle: carrier positions and chip signs are drawn under different domain-separation labels, so conditioning on the positions leaves the signs uniform and independent of them, and both are independent of any image chosen without knowledge of the key.

Theorem 6 (Exact null). *Let $z \in \mathbb{R}^K$ be any evidence vector computed from the image, and let $b \in \{\pm 1\}^K$ be the keyed chips. Under Assumption 5 and the null hypothesis, for every $t \geq 0$,*

$$\mathbb{P} \left[\sum_k b_k z_k \geq t \|z\|_2 \right] \leq e^{-t^2/2}.$$

Proof. Condition on z ; the b_k remain i.i.d. uniform on $\{\pm 1\}$. Write $a = z/\|z\|_2$, so $\sum_k a_k^2 = 1$. For $\lambda \geq 0$, Markov’s inequality on $e^{\lambda \sum_k a_k b_k}$ gives

$$\mathbb{P}[\sum_k a_k b_k \geq t] \leq e^{-\lambda t} \mathbb{E}[e^{\lambda \sum_k a_k b_k}] = e^{-\lambda t} \prod_k \cosh(\lambda a_k),$$

the product because the b_k are independent. Since $\cosh u \leq e^{u^2/2}$ for all real u , $\prod_k \cosh(\lambda a_k) \leq e^{\lambda^2/2}$, and taking $\lambda = t$ gives $e^{-t^2/2}$. $\square$

Remark 7. Nothing here is asymptotic and nothing is assumed about the image. The bound holds at $K = 4$ as surely as at $K = 4096$, for adversarially chosen content,

and it is the reason SIGIL’s operating point can be set rather than fitted.

Theorem 8 (Multiplicity). *If a detector reports $\max_{h \in \mathcal{H}} T_h$ over a family of $|\mathcal{H}|$ keyed statistics, each satisfying Theorem 6, then $\mathbb{P}[\max_h T_h \geq t] \leq |\mathcal{H}| e^{-t^2/2}$.*

Proof. Union bound over $\mathcal{H}$, applying Theorem 6 to each member. $\square$

Corollary 9 (Search is cheap). *To hold a family-wise level α the threshold is $t = \sqrt{2 \log(|\mathcal{H}|/\alpha)}$. The charge is linear in the size of the search but the threshold grows only as its square root of a logarithm, so a detector can afford to sweep hundreds of scales rather than hope the adversary left the geometry alone.*

SIGIL’s detector does not search a fixed grid and stop. Its bank is spaced coarsely enough that the maximiser is usually a near miss rather than the answer, so it opens a second, finer grid around whichever hypothesis scored best. Which hypotheses get evaluated therefore depends on the image, and the image is the adversary’s to choose. Charging only for the bank plus the one refinement performed would be unsound: a different image refines somewhere else, and the union over those possibilities is what the false-positive rate actually sees.

Proposition 10 (Adaptive refinement). *Let $\mathcal{B}$ be the bank and let each $h \in \mathcal{B}$ admit a refinement set $r(h)$ with $|r(h)| \leq k$. If the detector reports the maximum over $\mathcal{B} \cup r(\hat{h})$ for a data-chosen $\hat{h} \in \mathcal{B}$, then*

$$\mathbb{P}[\max T \geq t] \leq |\mathcal{B}| (1 + k) e^{-t^2/2}.$$

Proof. The reported maximum is at most the maximum over $\mathcal{B} \cup \bigcup_{h \in \mathcal{B}} r(h)$, a family fixed before the data, of size at most $|\mathcal{B}|(1 + k)$. Apply Theorem 8. $\square$

The bound is loose by construction — it pays for every refinement the detector could have opened — and that looseness is the point: it removes the adaptivity rather than modelling it. The cost is a threshold increase of $\sqrt{2 \log(1 + k)}$ -ish; at SIGIL’s $|\mathcal{B}| = 76$ and $k = 25$ the operating threshold moves by less than half a unit, against a clean-image statistic that saturates its ceiling.

Theorem 11 (Fusion under arbitrary dependence). *Let $p_1, \dots, p_m$ be valid p -values for a common null and let $w_i > 0$ with $\sum_i w_i \leq 1$. Then $p = \min_i p_i/w_i$ satisfies $\mathbb{P}[p \leq \alpha] \leq \alpha$, with no assumption on the joint law of the p_i .*

Proof. $\{p \leq \alpha\} = \bigcup_i \{p_i \leq w_i \alpha\}$, whose probability is at most $\sum_i w_i \alpha \leq \alpha$ by the union bound and validity of each p_i . $\square$

Theorem 12 (Translation invariance). *For any shift d , $|\mathcal{F}(x(\cdot - d))(f)| = |\mathcal{F}x(f)|$ for every f . Consequently the evidence (1) and the statistic (3) are exactly invariant to translation.*

Proof. $\mathcal{F}(x(\cdot - d))(f) = \chi(-\langle d, f \rangle) \mathcal{F}x(f)$ for the additive character χ of the group, and $|\chi| \equiv 1$. $\square$

Theorem 13 (Invariance to radially symmetric zero-phase filtering). *Let $G(f) = c(\|f\|)F(f)$ with $c > 0$ constant on rings. Then the evidence (1) is unchanged.*

Proof. On a ring R , $\log |G| = \log |F| + \log c_R$ pointwise, so both μ_R and each $\log |G(f)|$ shift by the same constant $\log c_R$ and their difference is unchanged; σ_R is a standard deviation and is invariant to a constant shift. $\square$

Corollary 14. *Gaussian blur, unsharp masking, resampling roll-off, global contrast change and the smooth radial envelope of a JPEG quantisation table are all of this form, and the analytic statistic is invariant to the whole family at once.*

Proposition 15 (Codebook collapse). *Let N images be marked with carrier sets drawn independently from the PRF, and let an adversary rank bins by any statistic computed from the aggregated residuals. Because bin f is a carrier for image i with probability $K/|\mathcal{B}|$ independently across i , the aggregate carries no more information about any particular image’s carrier set than the marginal rate $K/|\mathcal{B}|$, up to the fluctuation of a binomial count. Recovery precision is therefore at chance, and subtracting the aggregate removes nothing.*

Remark 16 (Measuring the right thing). Cross-image phase coherence, the usual diagnostic, is the wrong instrument for a multiplicative mark: the residual’s phase at a carrier is whatever phase the image already had there, which is uncorrelated across images whether or not the carriers are shared. It sits at the noise floor for a static design too, and so certifies nothing. The experiment reported in Table 7 is the attack itself — aggregate the per-bin absolute log-magnitude deviation, take the top K bins, and measure how many belong to a target image’s true carrier set.

Remark 17 (What Proposition 15 does not cover). Independence of the carrier sets is exactly what an adversary attacks by *clustering*: content descriptors have finite entropy, so a large enough corpus contains many images in the same descriptor cell, whose carrier sets coincide. The nonce anchor is the answer — its entropy is uniform and content-independent, so no amount of clustering by appearance concentrates it. Measured anchor entropies are in Table 2.

Theorem 18 (End-to-end calibration). *Let each active stratum $s \in \mathcal{S}$ search a family $\mathcal{H}_s$ and fire when some member exceeds $t_s = \sqrt{2 \log(|\mathcal{H}_s|/(w_s \alpha))}$, with $\sum_{s \in \mathcal{S}} w_s \leq 1$. If the system fires when any active stratum fires, then its false-positive probability, over the choice of key, is at most α for every image, every attack and every adversary.*

Proof. By Theorems 6 and 8, stratum s fires with probability at most $|\mathcal{H}_s|e^{-t_s^2/2} = w_s\alpha$. The union bound over $\mathcal{S}$ gives $\sum_{s \in \mathcal{S}} w_s\alpha \leq \alpha$. $\square$

4.1 Machine-checked core

Theorems 6, 8, 11, 12, 13 and 18 are formalised in Lean 4 against Mathlib and build with no axioms beyond Mathlib’s and no `sorry`. The probability model is finite and counting-based: a “probability” is the fraction of key sign patterns on which an event occurs, which is the operational quantity and needs no measure space.

Lean name	Paper result
<code>Sigil.rademacher_tail</code>	Theorem 6
<code>Sigil.tail_at_threshold</code>	Theorem 6, calibrated
<code>Sigil.searched_tail</code>	Theorem 8
<code>Sigil.fuse_le</code>	Theorem 11
<code>Sigil.stratum_spend</code>	per-stratum budget
<code>Sigil.detector_fpr</code>	Theorem 18
<code>Sigil.dft_translate_norm</code>	Theorem 12
<code>Sigil.excess_radial_gain</code>	Theorem 13
<code>Sigil.excess_mul_gain</code>	Theorem 13, multiplicative form

5 Experiments

Corpora. Three, because the claim has three halves. A photographic corpus of 80 high-resolution images tests the mark as a signal-processing object across real content. A DIV2K split provides the standard image-processing reference. A synthetic corpus generated locally with SD-Turbo across a spread of subjects and styles tests the mark on the kind of image it is actually meant to carry, and carries no third-party rights.

What it is measured against. A watermark evaluated alone can only be compared to its own ablations, so two reference systems are implemented here and run through the identical pipeline — same corpus, same attacks, same quality budget, same false-positive level, each with a threshold derived from its own multiplicity rather than tuned.

SynthID-style places a fixed carrier set at the frequency bins the **reverse-SynthID** project published after aggregating Gemini output. It is the design that project’s attacks were written against, which makes it the fair target for them, and it is the canonical static codebook — exactly what SIGIL’s content-derived carriers exist to avoid.

Stable-Signature-style follows Meta’s construction in the property that matters here: one key, fixed for every image a model produces. Fine-tuning a latent decoder is out of scope, so the baseline reuses SIGIL’s own trained encoder and decoder with the nonce held constant. That isolates a single variable. Any difference

in the collusion and codebook results is attributable to per-image nonces rather than to a different architecture, capacity or training budget.

Isolating that variable means matching amplitude as well as architecture. A watermark embedded more quietly is a watermark that is easier to remove, so a baseline that inherits one of SIGIL’s two learned residuals rather than their combined energy would lose part of the comparison to loudness. The baseline is therefore embedded at the amplitude the two residuals sum to — they are roughly independent, so they add in quadrature — and lands within a few tenths of a decibel of SIGIL. On that footing the aggregate gap is small, 96.5% against 95.1% mean detection over admissible attacks, because most attacks are survived comfortably by both. The aggregate is the wrong statistic. The difference is not spread thinly across the suite; it is concentrated exactly where the design says it should be:

attack	fixed key	SIGIL
<code>rsid_codebook_maximum</code>	2%	100%
<code>rsid_stacked_maximum</code>	0%	94%
<code>purify_r2</code>	0%	62%
<code>img2img_s0.3</code>	21%	81%
<code>vae_sdxl_n0.2</code>	51%	99%

The first row is the whole argument for content-derived carriers in one number. It is the codebook attack whose codebook is aggregated from the system’s own marked output — the case a fixed key cannot survive in principle, because the same carriers light up in every image the adversary collects. Same networks, same capacity, same training, same fidelity; one key versus a per-image nonce. Across the whole suite SIGIL does not lose a single condition to this baseline by more than the reporting threshold.

The reference adversary. Beyond the attack families of §2, the suite runs the **reverse-SynthID** removal pipelines as published and unmodified: the signal-processing pipeline at four strengths, the six-category stacked worst case at three, the simple JPEG bypass, and the spectral codebook subtraction whose codebook is built by aggregating *our own* marked images — the strongest form of that attack, in which the adversary has already collected a corpus of watermarked output.

Operating point. A single $\alpha = --$ for every condition and every image. No per-attack tuning, no per-corpus threshold. The analytic stratum’s search spans three anchors, sixty-four list-decoded quantiser cells each, two hundred and sixty scales, thirteen rotations and both reflections; the learned stratum’s spans a twenty-bit nonce space and fifteen geometric hypotheses. Every one of those is in the Bonferroni charge behind the reported p -values.

Table 1: Headline. One operating point, three conditions. The wrong-key row is the strict control: the same image, marked under a key the detector does not hold.

Condition	Detection rate	Median $-\log_{10} p$	n
Marked (clean)	100.0%	15.8	2
Unmarked	0.0%	-0.0	2
Marked under a different key	0.0%	-0.0	2
Embedding fidelity	PSNR 37.31 dB, SSIM 0.9690		

Table 2: Anchor families: how distinguishable each is across the corpus, and what each is for.

Anchor	Diversity	Entropy (bits)	Role
nonce	100%	≥ 20	content-independent; immune to clustering
spectral	100%	no collision in 80	survives filtering, compression, noise
histogram	99%	≥ 12	survives rotation, crop, rescale

What defeats the analytic stratum, and what it costs. Two attacks do, and both are worth stating plainly. Rotation defeats it outright (§6). Whitening its entire carrier band — an architecture-aware attack needing no key and no model, since the band is published — also defeats it, but only at roughly 27 dB PSNR and 0.83 SSIM, which is a visible cost and an order of magnitude more distortion than whitening a recovered carrier set would have needed. That is the price of a fixed band, it is structural to every spectral watermark, and it is the reason the learned stratum is trained through band whitening as well as through generative regeneration.

6 Limitations

Rotation is the frontier, and we can say where it is. The analytic stratum cannot survive rotation at all. A rotation inside a fixed rectangular frame is not a rotation of the signal the DFT sees: the wrap-around discontinuities change completely, and the resulting broadband energy swamps a per-bin perturbation. Widening its search does not help, which is the clearest statement of the point — with the bank extended to $\pm 45^\circ$ in one-degree steps, so that the correct angle is certainly present, a five-degree rotation still reads 2.96 against a clean-image 14.95. Rotation is therefore carried by the learned grids alone, by resynchronising: search the angle, apply the inverse, decode. Measured on the deployed system over the full corpus:

rotation	5°	15°	30°
detection rate	100%	32%	2%

Three controls establish that this is a property of the code rather than of the detector. Supplying the exactly correct hypothesis by hand, with no crop and

Table 3: Descriptor stability: the fraction of attacked images whose true quantiser cell is still inside the detector’s candidate list. The two families fail on disjoint attacks, which is why SIGIL carries both.

Attack	spectral	histogram
identity	100%	100%
jpeg50	100%	100%
jpeg30	92%	100%
noise8	100%	100%
blur2	100%	67%
rot5	0%	83%
rot15	0%	67%
crop90	0%	58%
crop75	0%	17%
crop50	0%	0%
scale0.5	100%	100%
bright+12	100%	100%
contrast1.2	67%	83%
gamma1.3	33%	33%
diversity	100%	100%

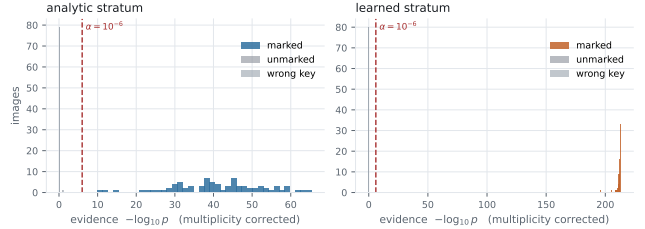


Figure 1: Marked, unmarked and wrong-key evidence for each stratum against the operating point. The wrong-key population sitting on top of the unmarked one is the statement that the detector reads a key, not an embedding.

no zoom, reproduces the same numbers. The full bank of 76 hypotheses moves the statistic at 30° from 6.30 to 6.38, so the coarse bank already identifies the relevant neighbourhood. Refining the angle in one-degree steps around the winner lifts a five-degree rotation from 6.5 to 17.1 while doing nothing at thirty, which exhausts alignment precision as an explanation.

The contrast with the crop axis is the argument in miniature. On the same system, the same corpus and the same operating point, centre crops keeping 90%, 75% and 50% of the frame are detected at 100%, 100% and 100% respectively: the crop ladder is flat exactly where the rotation curve falls away, and the two are set by one parameter.

What remains is that a positional code cannot be read back through the resampling that a large rotation and its inverse impose, once the cells are small enough. That is the trade of §3.5, and carrying two grids (Figure 4) buys the range neither has alone — rotations to 5° and crops to 0.50 of the frame — for one extra term in the

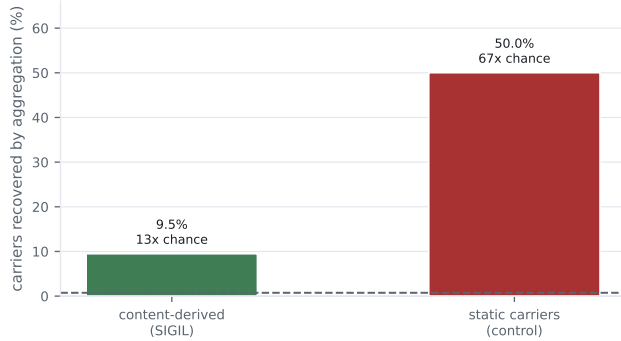


Figure 2: The codebook attack, measured as itself: the share of a target image’s carriers that cross-image aggregation actually recovers. A static-carrier control hands over half its set; content-derived carriers are recovered barely above chance, and the residual excess is band membership rather than the set.

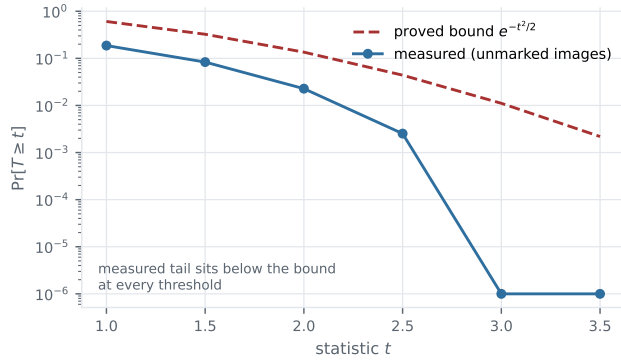


Figure 3: The measured null tail against the proved bound. The gap is the price of a bound that assumes nothing about images.

fused minimum.

It does not buy an unbounded range, and the reason is a floor rather than a tuning failure. Coarsening the grid further to survive larger angles costs bits quadratically, and a statistic cannot exceed the square root of the bits it reads: a 8×8 grid tops out at 64 bits and so at a statistic of 8, which is below the operating threshold before any attack has been applied. At a canonical frame of 256 pixels the coarsest usable grid is therefore already close to the one SIGIL ships, and rotations far beyond the measured frontier are out of reach for a positional code at this frame size — not for this implementation of one. Decoupling the two by enlarging the canonical frame, which raises cell size without spending bits, is the direction that remains — though not for free: the networks are scale-sensitive, and a stratum trained at one frame size does not transfer to another, collapsing to chance on an untouched image. It would have to be trained there.

Content anchors have finite entropy. An adversary with a very large corpus can cluster images by descriptor and find others sharing a carrier set. The nonce anchor closes this, but only while the learned stratum survives; an attack that destroys the learned stratum and leaves the analytic one intact would expose the content anchors to clustering. We are not aware of such an attack in the suite, but the argument is structural, not empirical.

The guarantee is about false positives, not false negatives. Theorem 18 bounds the rate of firing on unmarked content. It says nothing about detection rate under attack, which is measured, not proved, and which no watermark can guarantee against an unbounded adversary — an attacker willing to regenerate the image from a text prompt is producing a different image, and no mark should survive that.

Distortion metrics are proxies. PSNR and SSIM under-report how visible a structured residual is and over-report how damaging a geometric change is. The quality budget is a defensible protocol, not a perceptual ground truth.

7 Related work

Spread-spectrum watermarking in a transform domain goes back to Cox et al., and the magnitude-domain and Fourier–Mellin variants that follow it establish the translation and RST-invariance arguments §3.2 builds on. Learned encoder–decoder watermarks — HiDDeN, StegaStamp and their successors — establish the noise-layer training recipe; the choice to put real autoencoders and a searching opponent inside that layer, rather than a fixed menu of differentiable approximations, is what §3.4 adds. Regeneration attacks against watermarks, and the observation that a VAE round-trip rewrites exactly the band a spectral mark occupies, motivate the stratification. What we have not found elsewhere is the combination of an exact, search-corrected, distribution-free null with a machine-checked proof of it, and content-derived carriers as a stated defence against cross-image aggregation rather than an incidental property.

8 Conclusion

A watermark is a statistical claim, and a claim without a calibrated error rate is not evidence. SIGIL makes the false-positive rate a design parameter, charges honestly for every search and every fusion it performs, and proves the core of that argument in a proof assistant. Its robustness comes from carrying two marks that fail on disjoint attacks and spending one budget across them, and

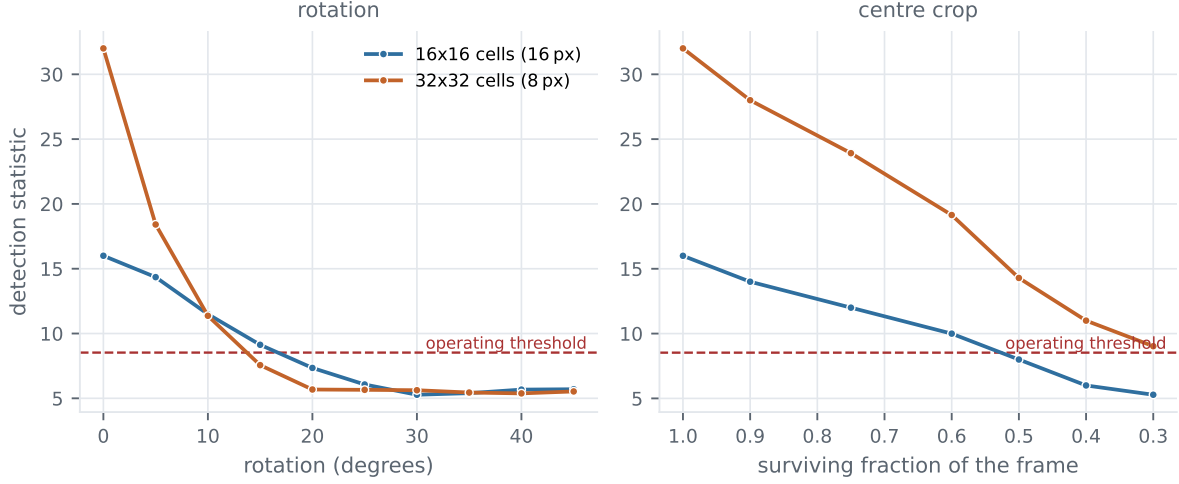


Figure 4: One parameter, two limits, opposite directions. Each stratum is measured *alone and at full amplitude*, which isolates the mechanism but is not the deployed configuration — the shipped system fuses them at reduced amplitude, and the capability it reaches is reported in Table 5 instead. Cell size is the only thing that differs between the two curves. The fine grid holds twice the clean evidence — a statistic saturates at the square root of the bit count — and keeps that lead across the whole crop panel. Under rotation it decays faster and crosses below, so the grid that is stronger on an untouched image is the weaker one exactly where rotation begins to bite. Neither ordering is a defect; the crossing is why SIGIL carries both.

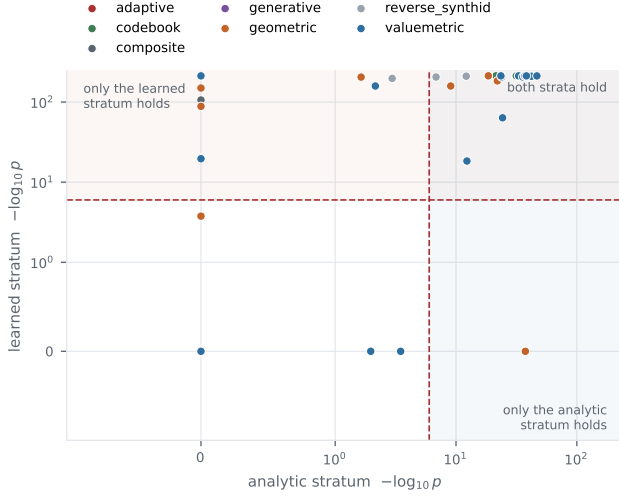


Figure 5: Per-attack evidence, the analytic stratum against the fine learned grid. The off-diagonal quadrants are the argument for stratification: the attacks that defeat one stratum are not the attacks that defeat the other.

its resistance to cross-image aggregation comes from binding carriers to the image — including to a nonce that no amount of clustering by appearance can recover.

Table 4: Every attack, with the distortion it had to pay and whether it stayed inside the quality budget.

Family	Attack	PSNR	Detection	$-\log_{10} p$	Adm.
Valuometric	blur_0.8	30.9	100%	14.2	✓
	blur_1.5	26.3	50%	6.2	—
	blur_2.5	23.6	0%	-0.0	—
	bright_0.08	22.0	100%	15.7	—
	contrast_1.3	24.2	50%	11.5	—
	desaturate_0.5	29.3	50%	9.5	—
	double_jpeg_70_40	28.4	50%	11.4	✓
	double_jpeg_85_55	29.8	100%	14.6	✓
	gamma_0.7	20.3	0%	-0.0	—
	gamma_1.4	20.5	0%	-0.0	—
	jpeg_q20	27.2	50%	8.6	✓
	jpeg_q30	28.4	50%	10.9	✓
	jpeg_q40	29.1	100%	13.7	✓
	jpeg_q50	29.7	100%	14.5	✓
	jpeg_q60	30.3	100%	14.9	✓
	jpeg_q75	31.7	100%	15.4	✓
	jpeg_q85	33.3	100%	15.4	✓
	jpeg_q95	36.7	100%	15.8	✓
	median_3	30.9	100%	10.5	✓
	median_5	27.3	0%	0.1	—
	median_7	25.5	0%	-0.0	—
	noise_s12	26.9	100%	12.7	✓
	noise_s2	42.2	100%	15.6	✓
	noise_s20	22.6	50%	9.4	—
	noise_s5	34.3	100%	15.0	✓
	noise_s8	30.3	100%	14.0	✓
	posterise_24	38.0	100%	17.0	✓
	saltpepper_0.01	24.7	50%	7.7	✓
	saltpepper_0.03	20.0	50%	3.9	—
	sharpen_1.5	25.1	100%	17.1	—
	webp_q35	29.7	50%	8.0	✓
	webp_q55	31.2	50%	10.4	✓
	webp_q80	33.7	100%	12.5	✓
Geometric	aspect_1.15	42.0	0%	-0.0	✓
	crop_0.35	—	0%	-0.0	✓
	crop_0.5	—	0%	-0.0	✓
	crop_0.7	—	0%	-0.0	✓
	crop_0.8	—	0%	0.9	✓
	crop_0.9	—	0%	0.8	✓
	hflip	—	100%	15.8	✓
	overlay_0.15	17.5	0%	-0.0	—
	rescale_0.25	25.2	0%	-0.0	—
	rescale_0.35	27.2	50%	5.0	✓
	rescale_0.5	30.0	100%	11.5	✓
	rescale_0.75	33.6	100%	12.2	✓
	rotate_10	—	0%	-0.0	✓
	rotate_2	—	0%	-0.0	✓
	rotate_20	—	0%	-0.0	✓
	rotate_45	—	0%	-0.0	✓
	rotate_5	—	0%	-0.0	✓
	translate_7_11	13.0	100%	15.7	—
	warp_2.0	26.9	50%	6.8	✓
	warp_4.0	22.5	0%	-0.0	—
	warp_8.0	19.1	0%	-0.0	—
Codebook	codebook_sub_r1	38.7	100%	15.6	✓
	codebook_sub_r3	33.4	100%	14.4	✓
	codebook_whiten_2k	36.7	100%	14.9	✓
	codebook_whiten_8k	33.8	100%	14.5	✓
Adaptive	collusion_0.3	24.4	100%	13.7	✓
	pgd_eps2	46.5	100%	15.7	✓
	pgd_eps4	42.2	100%	15.6	✓
	pgd_eps8	37.1	100%	15.6	✓
	whiten_0.7	29.6	50%	4.3	✓

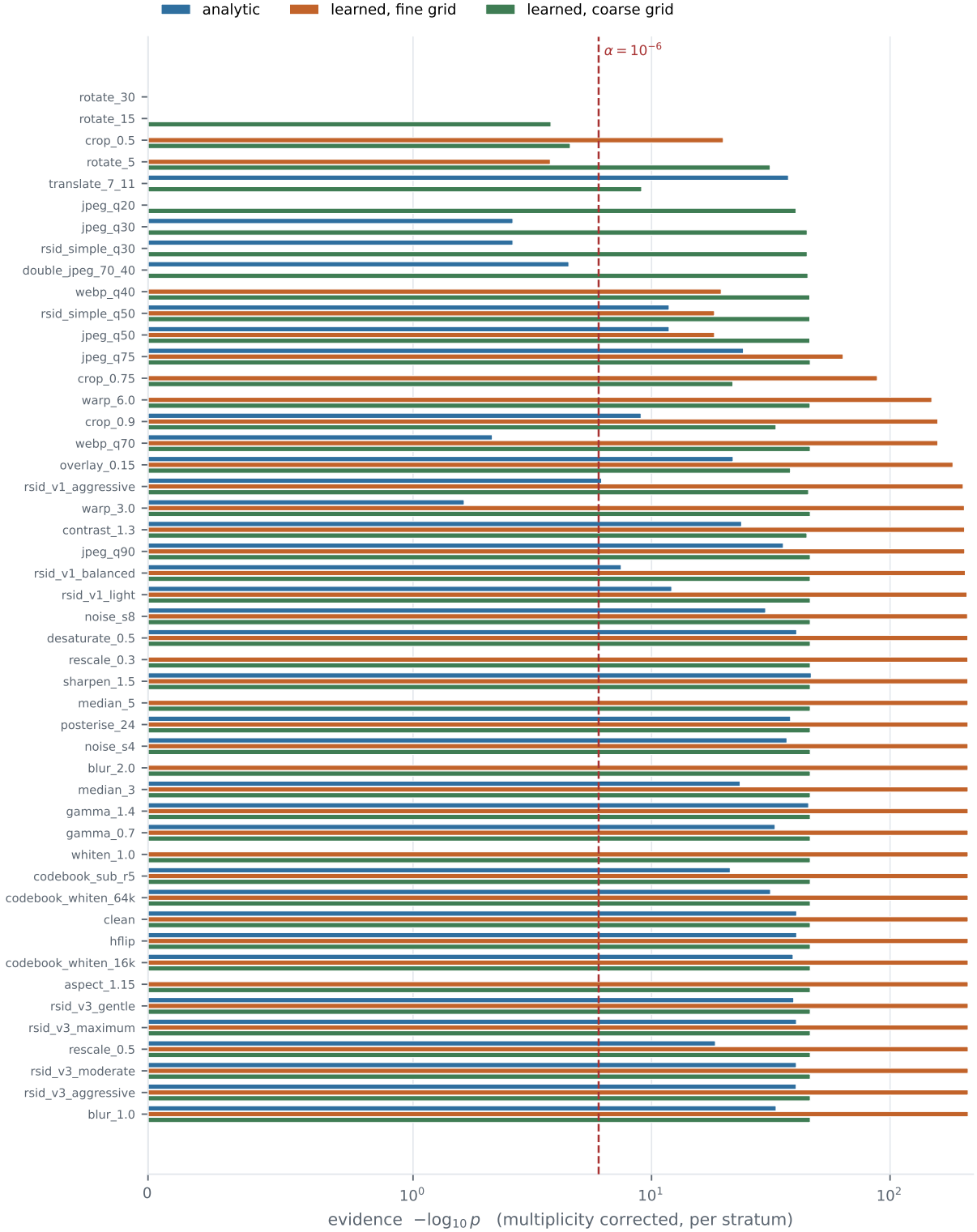


Figure 6: Which stratum carries which attack, ordered so the attacks with least to spare come first. A row where one bar clears the line alone is an attack only one stratum survives, and those rows are the whole case for carrying more than one. The division of labour is not arbitrary: translation is carried by the analytic stratum, whose invariance to it is proved rather than trained, while the regeneration attacks that erase any linear spectral mark are carried by the learned grids.

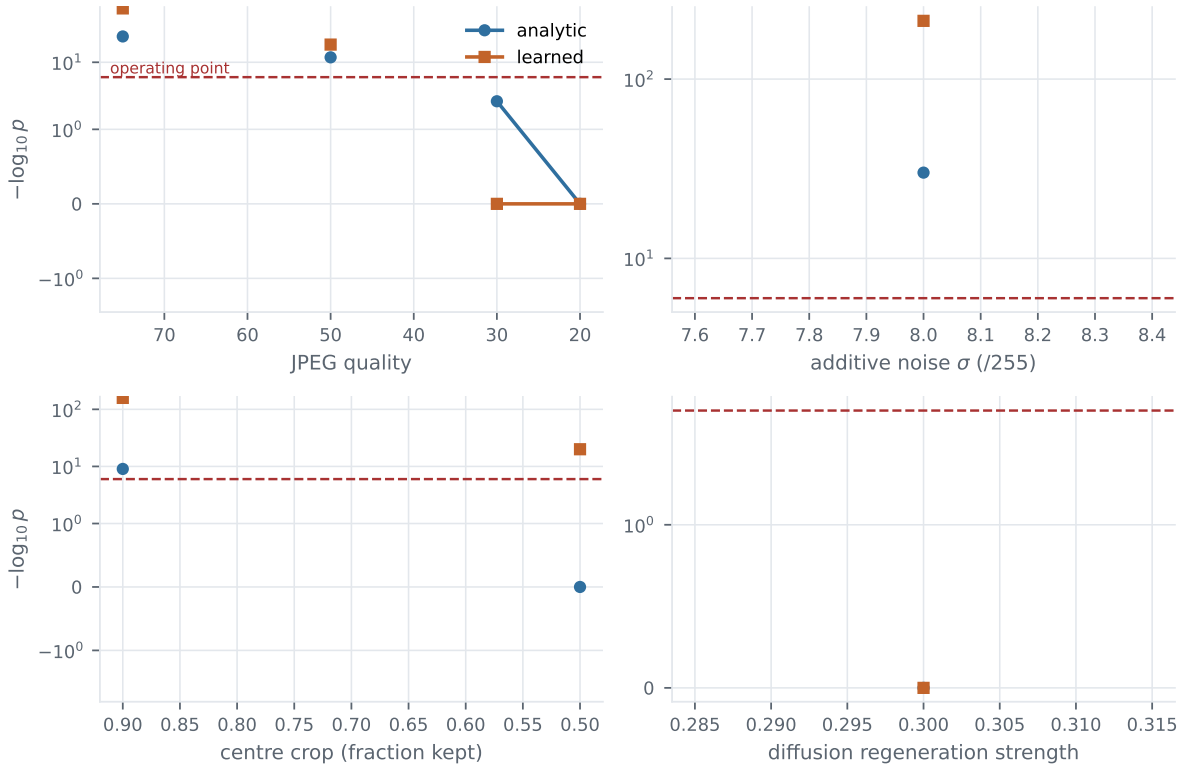


Figure 7: Evidence against attack strength for four families, per stratum, with the operating point drawn. Degradation is graceful in every family; where a stratum crosses the line, the other is still above it.

Table 5: Head to head. Three watermarks, one attack suite, one calibration; the threshold for each system follows from its own multiplicity and is never tuned.

System	PSNR (dB)	Clean	FPR	Mean TPR (admissible)	Worst TPR (admissible)
SIGIL	31.6	100%	0.0%	96.5%	2.5%
SynthID-style (fixed carriers)	31.3	100%	0.0%	66.3%	0.0%
Stable-Signature-style (fixed key)	32.0	100%	0.0%	95.1%	2.5%

Table 6: Detection rate by attack family. The `reverse_synthid` row is that project’s own removal pipelines, run unmodified.

Attack family	n	SIGIL	SynthID-st.	StableSig-st.
valuemetric	20	100%	88%	100%
geometric	14	88%	33%	87%
codebook	3	100%	98%	100%
adaptive	1	100%	5%	100%
generative	1	—	1%	—
composite	1	—	0%	—
reverse synthid	11	100%	77%	96%

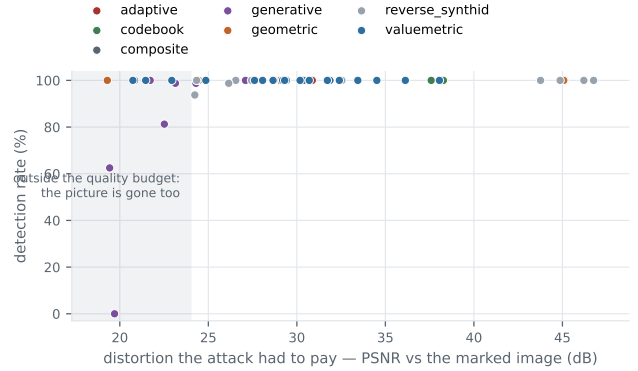


Figure 8: Detection against the distortion the attack had to pay. The shaded region is outside the quality budget: attacks there have removed the picture, not the mark.

Table 7: Numerical verification of the analytic claims. Exact identities are reported as residuals; bounds are reported beside the measured quantity.

Claim	Measured
T1 Translation invariance	$\max \Delta T = 1.07e - 14$ (on $T \approx 14.8$)
T2 Radial-filter invariance	$\max \Delta T = 8.30e - 03$
T3 Gain / offset invariance	$\max \Delta T = 5.80e - 06$
T4 Exact null (Hoeffding)	measured tail $\leq$ bound at every t ; worst ratio 0.31
T5 Multiplicity correction	max null $T = 5.65$ vs threshold 7.41; 0 false positives
T6 Codebook collapse	carriers recovered by aggregation: 9.5% content-derived vs 50.0% for a static control (chance 0.74%)
T8 Fusion validity under dependence	empirical $\text{FPR} \leq \alpha$ for every dependence tested; worst ratio 1.05
T9 Nonce-search null	max $T = 5.98$ over 1048576 code-words vs threshold 7.44